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Quadrature divider error correction

Abstract

A circuit including a frequency divider configured to receive a plurality of first frequency input clock signals and provide a plurality of second frequency output clock signals, wherein the plurality of second frequency output clock signals are lower in frequency than the plurality of first frequency input clock signals, a phase detector configured to determine a difference between the plurality of second frequency output clock signals, a low pass filter configured to measure a clock signal spacing error associated with the plurality of second frequency output clock signals based on the difference between the plurality of second frequency output clock signals and to generate one or more control signals in response to the clock signal spacing error, and a control unit configured to generate one or more corrected first frequency input clock signals based on the one or more control signals.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) This disclosure generally relates to systems for and methods of frequency division, including but not limited to systems and methods for providing clock signals at different frequencies and synchronizing multiple frequency clock dividers.

BACKGROUND OF THE DISCLOSURE

(2) In the last few decades, the market for electronic devices has grown by orders of magnitude, fueled by the use of portable devices, and increased connectivity, data transfer, and data storage in all manners of devices. Many modern electronic devices rely upon clock signals. A clock signal may be defined as a signal that oscillates between a high and low state at a constant frequency and is used to synchronize the actions of one or more electric circuits and devices in some embodiments. For example, systems on a chip (SoCs) often use several clocks domains with different programmable frequencies to realize a wide range of functionalities and cover different modes of operation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various objects, aspects, features, and advantages of the disclosure will become more apparent and better understood by referring to the detailed description taken in conjunction with the accompanying drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

(2) FIG. 1A is a block diagram depicting an embodiment of a computing environment including one or more access points in communication with one or more wireless devices or stations;

- (3) FIGS. 1B and 1C are block diagrams depicting embodiments of computing devices useful in connection with the methods and systems described herein.
- (4) FIG. 2 is a block diagram depicting an exemplary system including a pair of fractional frequency dividers, according to one or more embodiments.
- (5) FIG. 3 includes two timing diagrams showing aligned clock signals for a quadrature frequency divider, according to one or more embodiment.
- (6) FIG. 4 is a timing diagram showing a first skewed clock signals for a quadrature frequency divider, according to one or more embodiments.
- (7) FIG. 5 is a timing diagram showing a second skewed clock signals for a quadrature frequency divider, according to one or more embodiments.
- (8) FIG. 6 is a schematic circuit diagram showing a quadrature frequency divider, according to one or more embodiments.
- (9) FIG. 7 is block diagram of an error correction system for the quadrature frequency divider of FIG. 6, according to one or more embodiments.
- (10) FIG. 8 is a first timing diagram showing clock signals associated with a phase detector of the error correction system of FIG. 7, according to one or more embodiments.
- (11) FIG. 9 is a second timing diagram showing clock signals associated with a phase detector of the error correction system of FIG. 7, according to one or more embodiments.
- (12) FIGS. 10A-10B show control signals for controlling the clock signals and the corresponding clock signals created by the control signals, according to one or more embodiments.
- (13) FIG. 11 is a schematic diagram showing multiple high-speed clocks undergoing frequency division to generate multiple sub-rate clocks, according to one or more embodiments.
- (14) FIG. 12 is a first timing diagram showing clocks for the multiple high-speed clocks and sub-rate clocks of FIG. 11, according to one or more embodiments.
- (15) FIG. 13A is a schematic diagram of a circuit for a slip divider used to synchronize multiple sub-rate clocks, according to one or more embodiments.
- (16) FIGS. 13B-13C is a waveform diagram showing the waveforms signals for a clock frequency divider synchronization technique, according to one or more embodiments.
- (17) FIG. 14 is a first circuit diagram showing a frequency divider synchronizer, according to one or more embodiments.
- (18) FIG. 15 is a second circuit diagram showing a frequency divider synchronizer, according to one or more embodiments.
- (19) FIG. 16 is a third circuit diagram showing a frequency divider synchronizer, according to one or more embodiments.
- (20) FIG. 17 is a fourth circuit diagram showing a frequency divider synchronizer, according to one or more embodiments.
- (21) The details of various embodiments of the methods and systems are set forth in the accompanying drawings and the description below.

DETAILED DESCRIPTION

- (22) The following IEEE standard(s), including any draft versions of such standard(s), are hereby incorporated herein by reference in their entirety and are made part of the present disclosure for all purposes IEEE 802.3, IEEE 802.11x, IEEE 802.11ad, IEEE 802.11ah, IEEE 802.11aj, IEEE 802.16 and 802.16a, and IEEE 802.11ac. In addition, although this disclosure may reference aspects of these standard(s), the disclosure is in no way limited by these standard(s).
- (23) For purposes of reading the description of the various embodiments below, the following descriptions of the sections of the specification and their respective contents may be helpful. Section A describes a network environment and computing environment which may use frequency division according to some embodiments described herein, and Sections B-D describe embodiments of systems for and method of frequency division. Although this disclosure can reference aspects of various standard(s) and specification(s), the disclosure is in no way limited to

these aspects.

(24) In some embodiments, systems and methods provide clock domains for integrated circuits (e.g., SoCs). The systems and methods can reduce area and power consumption overhead associated with phase locked loop (PLL) cores that are used to generate the clock signals. In some embodiments, systems and methods meet frequency resolution, range, and low-jitter requirements that are very difficult to realize using conventional PLL cores. In some embodiments, a fractional divider is provided in silicon products, application specific circuits (ASIC), digital signal processors, programmable logic circuits, SoCs, or standard products that need highly programmable, low-jitter clock generators. In some embodiments, a fractional divider provides very fine ratio-based frequency resolution, uses small silicon area, provides low-jitter performance, provides unlimited frequency modulation capability, and is scalable and portable to newer integrated circuit processes. In some embodiments, systems and methods of fractional frequency division are used in modern high-speed and power-efficient data communication and processing systems, including but not limited to any backplane, serializer/deserializer (SERDES), Ethernet physical layer (PHY), optical transceiver, digital signal processor (DSP) module in coherent transceivers, input/output (I/O) interfaces, multicore processors, memories, power management, and wireless transceivers.

(25) Some embodiments relate to a frequency divider for providing a lower frequency clock output based on a higher frequency clock input. In some embodiments, the frequency divider may produce output clock signals with clock signal spacing error. The systems and methods described herein provide a technique for correcting the clock signal spacing error.

(26) Further, some embodiments relate to synchronizing sub-rate clocks which may be used in communication systems. In many analog to digital converters (ADCs), deserializer circuits in link communication (SERDES), or even radio receivers (RF radios), the information is processed synchronously with a clock. As the data rates have increased to well above 20 Gb/s, it is generally preferred to process the data with multiple sub-rate clocks. For example, in one embodiment, a data rate of 20 Gb/s can be captured with two 10 GHz clocks, or four 5 GHz clocks, etc. When using such sub-rate clocks in communication systems with a certain data rate (e.g., 20 Gb/s), it may be desired that the multiple sub-rate clocks are as equally spaced in time as possible so that the timing reference is equal to the full data rate itself. For example, in the case of a 20 Gb/s system, the timing reference would be at 20 GHz, or a period of 50 ps. In such a case, the data rate may be implemented with a sub-rate system with four 5 GHz clocks, where ideally the four clocks' rising edges (timing references) are spaced by exactly 50 ps. A rising edge may refer to when a clock signal moves from a low state (e.g., 0) to a high state (e.g., 1) in some embodiments. To ensure this, the sub-rate clocks may be synchronized when dividing the frequency to create the sub-rate clocks. A sub-rate clock signal may refer to an output signal of a frequency divider which is at a lower frequency than the input to the frequency divider in some embodiments. The systems and methods described herein provide a technique for providing such a synchronization.

(27) A. Computing and Network Environment

(28) Prior to discussing specific embodiments of the present solution, it may be helpful to describe aspects of the operating environment as well as associated system components (e.g., hardware elements) in connection with the methods and systems described herein. Referring to FIG. 1A, an embodiment of a network environment is depicted. The network may include or be in communication with one or more storage area networks (SANs), security adapters, or Ethernet converged network adapters (CNAs). In brief overview, the network environment includes a wireless communication system that includes one or more access points **106**, one or more wireless communication devices **102** and a network hardware component **192**. The wireless communication devices **102** may for example include laptop computers **102**, tablets **102**, personal computers **102**, wearable devices **102**, vehicles **102** (e.g., automobiles, drones, smart vehicles, robotic units, etc.), smart televisions **102**, gaming consoles **102**, internet of things (IoT) devices **102**, cellular telephone

devices **102**, and/or any other electronic devices capable of wireless communication. The details of an embodiment of wireless communication devices **102** and/or access point **106** are described in greater detail with reference to FIGS. **1B** and **1C**. The network environment can be an ad hoc network environment, an infrastructure wireless network environment, a wired network coupled to a wireless network, a subnet environment, etc., or a combination of the foregoing, in one embodiment.

(29) The access points (APs) **106** may be operably coupled to the network hardware **192** via local area network connections. The network hardware **192**, which may include one or more routers, gateways, switches, bridges, modems, system controllers, appliances, etc., may provide a local area network connection for the communication system. Each of the access points **106** may have an associated antenna or an antenna array to communicate with the wireless communication devices **102** in its area. The wireless communication devices **102** may register with a particular access point **106** to receive services from the communication system (e.g., via a SU-MIMO or MU-MIMO configuration). For direct connections (i.e., point-to-point communications), some wireless communication devices **102** may communicate directly via an allocated channel and communications protocol. Some of the wireless communication devices **102** may be mobile or relatively static with respect to the access point **106**.

(30) In some embodiments, an access point **106** includes a device or module (including a combination of hardware and software) that allows wireless communication devices **102** to connect to a wired network using Wi-Fi or other standards. An access point **106** may sometimes be referred to as a wireless access point (WAP). An access point **106** may be configured, designed and/or built for operating in a wireless local area network (WLAN). An access point **106** may connect to a router (e.g., via a wired network) as a standalone device in some embodiments. In other embodiments, an access point **106** can be a component of a router. An access point **106** can provide multiple devices access to a network. An access point **106** may, for example, connect to a wired Ethernet connection and provide wireless connections using radio frequency links for other wireless communication devices **102** to utilize that wired connection. An access point **106** may be built and/or configured to support a standard for sending and receiving data using one or more radio frequencies. Those standards and the frequencies they use may be defined by the IEEE (e.g., IEEE 802.11 standards). An access point **106** may be configured and/or used to support public internet hotspots, and/or on an internal network to extend the network's Wi-Fi signal range.

(31) In some embodiments, the access points **106** may be used for in-home or in-building wireless networks (e.g., IEEE 802.11, Bluetooth, ZigBee, or any other type of radio frequency based network protocol and/or variations thereof). Each of the wireless communication devices **102** may include a built-in radio and/or is coupled to a radio. Such wireless communication devices **102** and/or access points **106** may operate in accordance with the various aspects of the disclosure as presented herein to enhance performance, reduce costs and/or size, and/or enhance broadband applications. Each wireless communication device **102** may have the capacity to function as a client node seeking access to resources (e.g., data, and connection to networked nodes such as servers) via one or more access points.

(32) The network connections may include any type and/or form of network and may include any of the following: a point-to-point network, a broadcast network, a telecommunications network, a data communication network, or a computer network. The topology of the network may be a bus, star, or ring network topology. The network may be of any such network topology as known to those ordinarily skilled in the art capable of supporting the operations described herein. In some embodiments, different types of data may be transmitted via different protocols. In other embodiments, the same types of data may be transmitted via different protocols.

(33) The communications device(s) **102** and access point(s) **106** may be deployed as and/or executed on any type and form of computing device, such as a computer, network device, or appliance capable of communicating on any type and form of network and performing the

operations described herein. FIGS. 1B and 1C depict block diagrams of a computing device **100** useful for practicing an embodiment of the wireless communication device **102** or access point **106**. As shown in FIGS. 1B and 1C, each computing device **100** includes a central processing unit **121** and a main memory unit **122**. As shown in FIG. 1B, a computing device **100** may include a storage device **128**, an installation device **116**, a network interface **118**, an I/O controller **123**, display devices **124a-101n**, a keyboard **126** and a pointing device **127**, such as a mouse. The storage device **128** may include, without limitation, an operating system and/or software. As shown in FIG. 1C, each computing device **100** may also include additional optional elements, such as a memory port **103**, a bridge **170**, one or more input/output devices **130a-130n** (generally referred to using reference numeral **130**), and a cache memory **140** in communication with the central processing unit **121**.

(34) The central processing unit **121** is any logic circuitry that responds to and processes instructions fetched from the main memory unit **122**. In many embodiments, the central processing unit **121** is provided by a microprocessor unit, such as: those manufactured by Intel Corporation of Mountain View, California; those manufactured by International Business Machines of White Plains, New York; or those manufactured by Advanced Micro Devices of Sunnyvale, California. The computing device **100** may be based on any of these processors, or any other processor capable of operating as described herein.

(35) Main memory unit **122** may be one or more memory chips capable of storing data and allowing any storage location to be accessed by the central processing unit **121**, such as any type or variant of Static random access memory (SRAM), Dynamic random access memory (DRAM), Ferroelectric RAM (FRAM), NAND Flash, or NOR Flash and Solid State Drives (SSD). The main memory **122** may be based on any of the above described memory chips, or any other available memory chips capable of operating as described herein. In the embodiment shown in FIG. 1B, the central processing unit **121** communicates with main memory **122** via a system bus **150** (described in more detail below). FIG. 1C depicts an embodiment of a computing device **100** in which the central processing unit **121** communicates directly with main memory **122** via a memory port **103**. For example, in FIG. 1C the main memory **122** may be DRAM.

(36) FIG. 1C depicts an embodiment in which the central processing unit **121** communicates directly with cache memory **140** via a secondary bus, sometimes referred to as a backside bus. In other embodiments, the central processing unit **121** communicates with cache memory **140** using the system bus **150**. Cache memory **140** typically has a faster response time than main memory **122** and is provided by, for example, SRAM, BSRAM, or EDRAM. In the embodiment shown in FIG. 1C, the central processing unit **121** communicates with various I/O devices **130** via a local system bus **150**. Various buses may be used to connect the central processing unit **121** to any of the I/O devices **130**, for example, a VESA VL bus, an ISA bus, an EISA bus, a MicroChannel Architecture (MCA) bus, a PCI bus, a PCI-X bus, a PCI-Express bus, or a NuBus. For embodiments in which the I/O device is a video display **124**, the central processing unit **121** may use an Advanced Graphics Port (AGP) to communicate with the video display **124**. FIG. 1C depicts an embodiment of a computer or computing device **100** in which the central processing unit **121** may communicate directly with I/O device **130b**, for example via HYPERTRANSPORT, RAPIDIO, or INFINIBAND communications technology. FIG. 1C also depicts an embodiment in which local busses and direct communication are mixed: the central processing unit **121** communicates with I/O device **130a** using a local interconnected bus while communicating with I/O device **130b** directly.

(37) A wide variety of I/O devices **130a-130n** may be present in the computing device **100**. Input devices include keyboards, mice, trackpads, trackballs, microphones, dials, touch pads, touch screens, and drawing tablets. Output devices include video displays, speakers, inkjet printers, laser printers, projectors and dye-sublimation printers. The I/O devices **130a-130n** may be controlled by an I/O controller **123** as shown in FIG. 1B. The I/O controller **123** may control one or more I/O devices **130a-130n** such as a keyboard **126** and a pointing device **127**, e.g., a mouse or optical pen.

Furthermore, an I/O device **130a-130n** may also provide storage and/or an installation medium **116** for the computing device **100**. In other embodiments, the computing device **100** may provide USB connections (not shown) to receive handheld USB storage devices such as the USB Flash Drive line of devices manufactured by Twintech Industry, Inc. of Los Alamitos, California.

(38) Referring again to FIG. 1B, the computing device **100** may support any suitable installation device **116**, such as a disk drive, a CD-ROM drive, a CD-R/RW drive, a DVD-ROM drive, a flash memory drive, tape drives of various formats, USB device, hard-drive, a network interface, or any other device suitable for installing software and programs. The computing device **100** may further include a storage device, such as one or more hard disk drives or redundant arrays of independent disks, for storing an operating system and other related software, and for storing application software programs such as any program or software **120** for implementing (e.g., software **120** configured and/or designed for) the systems and methods described herein. Optionally, any of the installation devices **116** could also be used as the storage device. Additionally, the operating system and the software **120** can be run from a bootable medium.

(39) Furthermore, the computing device **100** may include a network interface **118** to interface to the network **104** through a variety of connections including, but not limited to, standard telephone lines, LAN or WAN links (e.g., 802.11, T1, T3, 56 kb, X.25, SNA, DECNET), broadband connections (e.g., ISDN, Frame Relay, ATM, Gigabit Ethernet, Ethernet-over-SONET), wireless connections, or some combination of any or all of the above. Connections can be established using a variety of communication protocols (e.g., TCP/IP, IPX, SPX, NetBIOS, Ethernet, ARCNET, SONET, SDH, Fiber Distributed Data Interface (FDDI), RS232, IEEE 802.11, IEEE 802.11a, IEEE 802.11b, IEEE 802.11g, IEEE 802.11n, IEEE 802.11ac, IEEE 802.11ad, CDMA, GSM, WiMax and direct asynchronous connections). In one embodiment, the computing device **100** communicates with other computing devices **100** via any type and/or form of gateway or tunneling protocol such as Secure Socket Layer (SSL) or Transport Layer Security (TLS). The network interface **118** may include a built-in network adapter, network interface card, PCMCIA network card, card bus network adapter, wireless network adapter, USB network adapter, modem or any other device suitable for interfacing the computing device **100** to any type of network capable of communication and performing the operations described herein.

(40) In some embodiments, the computing device **100** may include or be connected to one or more display devices **124a-124n**. As such, any of the I/O devices **130a-130n** and/or the I/O controller **123** may include any type and/or form of suitable hardware, software, or combination of hardware and software to support, enable or provide for the connection and use of the display device(s) **124a-124n** by the computing device **100**. For example, the computing device **100** may include any type and/or form of video adapter, video card, driver, and/or library to interface, communicate, connect or otherwise use the display device(s) **124a-124n**. In one embodiment, a video adapter may include multiple connectors to interface to the display device(s) **124a-124n**. In other embodiments, the computing device **100** may include multiple video adapters, with each video adapter connected to the display device(s) **124a-124n**. In some embodiments, any portion of the operating system of the computing device **100** may be configured for using multiple displays **124a-124n**. One ordinarily skilled in the art will recognize and appreciate the various ways and embodiments that a computing device **100** may be configured to have one or more display devices **124a-124n**.

(41) In further embodiments, an I/O device **130** may be a bridge between the system bus **150** and an external communication bus, such as a USB bus, an Apple Desktop Bus, an RS-232 serial connection, a SCSI bus, a FireWire bus, a FireWire 800 bus, an Ethernet bus, an AppleTalk bus, a Gigabit Ethernet bus, an Asynchronous Transfer Mode bus, a FibreChannel bus, a Serial Attached small computer system interface bus, a USB connection, or an HDMI bus.

(42) A computing device or system **100** of the sort depicted in FIGS. 1B and 1C may operate under the control of an operating system, which controls scheduling of tasks and access to system resources. The computing device **100** can be running any operating system such as any of the

versions of the MICROSOFT WINDOWS operating systems, the different releases of the Unix and Linux operating systems, any version of the MAC OS for Apple computers, any embedded operating system, any real-time operating system, any open source operating system, any proprietary operating system, any operating systems for mobile computing devices, or any other operating system capable of running on the computing device and performing the operations described herein. Typical operating systems include, but are not limited to: Android, produced by Google Inc.; WINDOWS 7 and 8, produced by Microsoft Corporation of Redmond, Washington; MAC OS, produced by Apple Computer of Cupertino, California; WebOS, produced by Research In Motion (RIM); OS/2, produced by International Business Machines of Armonk, New York; and Linux, a freely-available operating system distributed by Caldera Corp. of Salt Lake City, Utah, or any type and/or form of a Unix operating system, among others.

(43) The computing device **100** can be any workstation, telephone, desktop computer, laptop or notebook computer, server, handheld computer, mobile telephone or other portable telecommunications device, media playing device, a gaming system, mobile computing device, or any other type and/or form of computing, telecommunications or media device that is capable of communication. The computing device **100** has sufficient processor power and memory capacity to perform the operations described herein.

(44) In some embodiments, the computing device **100** may have different processors, operating systems, and input devices consistent with the device. For example, in one embodiment, the computing device **100** is a smart phone, mobile device, tablet or personal digital assistant. In still other embodiments, the computing device **100** is an Android-based mobile device, an iPhone smart phone manufactured by Apple Computer of Cupertino, California, or a Blackberry or WebOS-based handheld device or smart phone, such as the devices manufactured by Research In Motion Limited. Moreover, the computing device **100** can be any workstation, desktop computer, laptop or notebook computer, server, handheld computer, mobile telephone, any other computer, or other form of computing or telecommunications device that is capable of communication and that has sufficient processor power and memory capacity to perform the operations described herein. Aspects of the operating environments and components described above will become apparent in the context of the systems and methods disclosed herein.

(45) B. Frequency Divider

(46) With reference to FIG. 2, a system **200** is a DSP SERDES interface for use in a complex SOC. Although FIG. 2 depicts a particular application, the frequency division systems and methods can be used in any electronic application. System **200** can be used in and/or with the systems described with reference to FIGS. 1A-C.

(47) In some embodiments, the system **200** includes a signal source **202**, an analog to digital converter/digital to analog converter (ADC/DAC) unit **204**, DSP/PHY unit **206**, a high speed SERDES unit **208**, a signal source **210**, a fractional divider **212**, and a fractional divider **214**. Units **204**, **206** and **208** can operate at different clock frequencies (e.g., driven by different clock signals) in some embodiments. In some embodiments, the fractional divider **212** and fractional divider **214** provide programmable levels of synchronous fractional frequency division for DSP/PHY unit **206** or any other type of electronic device.

(48) Signal source **202** is a crystal integrated circuit that provides a first reference frequency signal at a first frequency in some embodiments. The first reference frequency signal is provided to the ADC/DAC unit **204**. Signal source **210** is a crystal integrated circuit that provides a second reference frequency signal at a second frequency in some embodiments. The second reference frequency signal is provided to the SERDES unit **208**.

(49) The ADC/DAC unit **204** is a high speed macro, unit, circuit, logic, or other implementation that can provide and receive data to and from DSP/PHY unit **206**. ADC/DAC unit **204** includes an integer or fractional N PLL **220** in some embodiments. PLL **220** receives the reference frequency signal from the signal source **202** and provides a clock signal at a first clock frequency. The first

clock signal can be a square wave or other signal for driving gates, processors or other logic provided at a frequency $F_{\text{sub.I/O1}}$ in some embodiments. The first clock signal can be a divided (e.g., integer or fractional) signal of the reference frequency signal and can be used by The ADC/DAC unit **204** to convert or otherwise process data.

(50) The DSP/PHY unit **206** processes data provided between units **204** and **208**. DSP/PHY unit **206** includes first in, first out (FIFO) memory **222** and FIFO memory **224**. FIFO memory **222** and FIFO memory **224** handle synchronization for communications between units **204** and **208** relying upon clock signals provided from fractional dividers **212** and **214**.

(51) The SERDES unit **208** is a high speed macro, unit, circuit, logic, or other implementation that can provide and receive data to and from DSP/PHY unit **206**. SERDES unit **208** includes an integer or fractional N PLL **226** in some embodiments. PLL **226** receives a reference frequency signal from the signal source **210** and provides a clock signal at a second clock frequency. The second clock signal can be a square wave or other signal for driving gates, processors or other logic provided at a frequency $F_{\text{sub.I/O2}}$ in some embodiments. The second clock signal can be a divided (e.g., integer or fractional) signal of the second reference frequency signal from signal source **210** and can be used by SERDES unit **208** to convert signals and data. The second clock signal can also be provided to fractional divider **214**.

(52) Fractional divider **212** is a macro, unit, circuit, logic, or other implementation that can provide a first divided clock signal at a selectable frequency using fractional division. In some embodiments, the fractional divider **212** divides the first clock signal by a divisor (e.g., the sum of an integer $P1$ and the quotient of an integer $Q1$ divided by $R1$) to provide the first divided signal at a frequency $F_{\text{sub.DSP1}}$. Fractional divider **214** is a macro, unit, circuit, logic, or other implementation that can provide a second divided clock signal at a selectable frequency using fractional division. In some embodiments, the fractional divider **214** divides the second clock signal by a divisor (e.g., the sum of an integer $P2$ and the quotient of an integer $Q2$ divided by $R2$) to provide the second divided signal at a frequency $F_{\text{sub.DSP2}}$. In some embodiments, fractional dividers **212** and **214** are standalone fractional divider (FDIV) macros. In some embodiments, the fractional dividers **212** and **214** may be quadrature frequency dividers which are configured to divide the reference signals by 2.

(53) Units **204** and **208** operate independently at different clock rates, and FIFO memories **222** and **224** handle the synchronization relying on precise ratios between clock frequencies ($F_{\text{sub.I/O1}}$, $F_{\text{sub.I/O2}}$, . . . etc.) and DSP clock frequencies ($F_{\text{sub.DSP1}}$, $F_{\text{sub.DSP2}}$, etc.) in some embodiments. Fractional dividers **212** and **214** advantageously do not employ multiple dividers and PLL cores to realize the desired ratios ($F_{\text{sub.DSP1}}/F_{\text{sub.I/O1}}=N_{\text{sub.1}}/M_{\text{sub.1}}$, $F_{\text{sub.DSP2}}/F_{\text{sub.I/O2}}=N_{\text{sub.2}}/M_{\text{sub.2}}$, etc.) where $N_{\text{sub.1}}$, $M_{\text{sub.1}}$, $N_{\text{sub.2}}$, and $M_{\text{sub.2}}$ are integers in some embodiments. Fractional dividers **212** and **214** advantageously do not occupy large silicon area and do not consume high power, yet meet tighter jitter specifications associated with the increased data rates in some embodiments.

(54) In some embodiments, fractional dividers **212** and **214** are reconfigurable and greatly simplify frequency planning in complex SoCs. In some embodiments, fractional dividers **212** and **214** reuse available high frequency clock signals from units **204** and **208** or share a single PLL core to generate multiple independent output clock signals. In some embodiments, fractional dividers **212** and **214** can achieve exact frequency ratios with very fine delta sigma ($\Delta\Sigma$) frequency resolution (up to 44 bit) for both numerator and denominator. In some embodiments, fractional dividers **212** and **214** are implemented in an all-digital circuit (e.g., all complementary metal oxide semiconductor (CMOS)) that is compact (e.g., one or two orders of magnitude smaller than a PLL core), scalable and easily portable to newer processes. In some embodiments, the fractional dividers **212** and **214** also achieve excellent low-jitter performance (<0.5 ps.sub.rms) across process-voltage-temperature (PVT) variations by using an adaptive background calibration technique. In some embodiments, fractional dividers **212** and **214** use an open loop architecture that

overcomes the bandwidth limitation of PLLs and achieves ideal spread spectrum modulation and instantaneous frequency switching without any frequency overshoot.

(55) C. Systems for and Methods for Frequency Division Error Correction

(56) With reference to FIG. 3, timing diagrams **300** and **302** showing aligned clock signals for a quadrature frequency divider are shown, according to an exemplary embodiment. A frequency divider may refer to a circuit which utilizes latches and/or flip flops and is configured to receive a higher frequency input clock signal and divide the frequency of the higher frequency input clock signal to produce a lower frequency output clock signal in some embodiments. The higher frequency input clock signal may refer to the input clock which has a higher frequency than the lower frequency output clock signal in some embodiments. Conversely, the lower frequency input clock signal may refer to the clock signal produced as an output of the frequency divider in some embodiments. A quadrature divider may refer to a type of frequency divider which receives two inputs and produces four outputs in some embodiments. A flip flop may refer a circuit element with two stable states that can be used to store binary data in some embodiments. A latch may refer to a circuit element that has two inputs and one output in some embodiments. Timing diagram **300** includes a first clock signal **304** and a second clock signal **306**. The first clock signal **304** is a high frequency clock signal “CK_P”. The second clock signal **306** is a complementary clock signal “CK_N” which has the opposite duty cycle as the first clock signal **304**. The first clock signal **304** and the second clock signal **306** may be used as input clocks to a clock frequency divider used to create quadrature clock signals included in the second timing diagram **302**. For example, in some embodiments, the first clock signal **304** may be used as signal source **202** while the second clock signal **306** may be used as the signal source **210**.

(57) The timing diagram **302** demonstrates the output clock signals **308**, **310**, **312**, and **314** from the quadrature divider. As can be seen, the frequency of the output clock signals **308**, **310**, **312**, and **314** is divided by 2. Further, as can be seen in timing diagram **302**, the time-spacing **316a-316d** between each of the clock signals are equally and ideally spaced. This type of spacing demonstrates an ideal quadrature divider without clock signal spacing error.

(58) With reference to FIG. 4, timing diagrams **400** and **402** showing clock signals for a quadrature frequency divider with clock signal spacing error are shown, according to an exemplary embodiment. A clock signal spacing error may refer to an output of a clock frequency divider where the timing of the rising edges is not equal between each of the output clock signals in some embodiments. The timing diagrams **400** and **402** are similar to timing diagrams **300** and **302** respectively except the clock signals shown in FIG. 4 are skewed. Timing diagram **400** includes a first clock signal **404** and a second clock signal **406**. The first clock signal **404** is a high frequency clock signal “CK_P”. The second clock signal **406** is a complementary clock signal “CK_N” which has the opposite duty cycle as the first clock signal **404**. As can be seen in FIG. 4, the timing of the second signal **406** is skewed by timing **405**. The first clock signal **404** and the second clock signal **406** may be used as input clocks to a clock frequency divider used to create quadrature clock signals included in the second timing diagram **402**. For example, in some embodiments, the first clock signal **404** may be used as signal source **202** while the second clock signal **406** may be used as the signal source **210**.

(59) The timing diagram **402** demonstrates the output clock signals **408**, **410**, **412**, and **414** from the quadrature divider. As can be seen, the frequency of the output clock signals **408**, **410**, **412**, and **414** is divided by 2. Further, as can be seen in timing diagram **402**, the time-spacing **416a-416d** between each of the clock signals are not equally and not ideally spaced. This type of spacing demonstrates a quadrature divider with clock signal spacing error.

(60) With reference to FIG. 5, timing diagrams **500** and **502** showing clock signals for a quadrature frequency divider with clock signal spacing error are shown, according to an exemplary embodiment. The timing diagrams **500** and **502** are similar to **500** and **502** respectively except the clock signals shown in FIG. 5 are skewed because the input signals have a duty cycle over 50%.

Timing diagram **500** includes a first clock signal **504** and a second clock signal **506**. The first clock signal **504** is a high frequency clock signal “CK_P”. The second clock signal **506** is a complementary clock signal “CK_N” which has the opposite duty cycle as the first clock signal **504**. As can be seen in FIG. 5, the duty cycle of each of the first clock signal **504** and the second clock signal **506** is higher than 50%. The first clock signal **504** and the second clock signal **506** may be used as input clocks to a clock frequency divider used to create quadrature clock signals included in the second timing diagram **502**. For example, in some embodiments, the first clock signal **504** may be used as signal source **202** while the second clock signal **506** may be used as the signal source **210**.

(61) The timing diagram **502** demonstrates the output clock signals **508**, **510**, **512**, and **514** from the quadrature divider. As can be seen, the frequency of the output clock signals **508**, **510**, **512**, and **514** is divided by 2. Further, as can be seen in timing diagram **502**, the time-spacing **516a-516d** between each of the clock signals are not equally and not ideally spaced because the duty cycles of the clock signals **504** and **506**. This type of spacing demonstrates a quadrature divider with clock signal spacing error.

(62) With reference to FIG. 6, a circuit diagram **600** showing a quadrature frequency divider **606** is shown, according to an exemplary embodiment. The circuit diagram **600** includes a flip flop **606** which is configured to divide the frequency of the input clock signals **602** and **604**. Specifically, the flip flop **606** may be divide-by-two flip flop which may be circuit component which is configured to receive an input signal and produce an output signal with half the frequency of the input signal. Specifically, the input clock signal **602** is fed into the frequency divider **606**. The divided frequency signal from frequency divider **606** is then fed into each of the latches **608**, **610**, **612**, and **614**. Each of the input clock signals **602** and **604** are used to alternatively clock the latches **608**, **610**, **612**, and **614** to create output frequency divided signals **616**, **618**, **620**, and **622**. In some embodiments, each of the latches **608**, **610**, **612**, and **614** may be the same. In one embodiment, the input clock signals **602** and **604** each have a duty cycle of 50% and no skew. In such a case, the output frequency signals **608**, **610**, **612**, and **614** will not have any clock signal spacing error. However, in a different embodiment, the input clock signals **602** and **604** may have a duty cycle above or below 50% and/or may have some skewed timing. In such a case, the quadrature frequency divider **606** may be used in conjunction with an error correction system which may create controllable input clock signals **602** and **604** in order to cancel the clock signal spacing error. The error correction system is explained in more detail with respect to FIG. 7 below.

(63) With reference to FIG. 7, an error correction system **700** is shown, according to an exemplary embodiment. The error correction system **700** includes the quadrature divider **606**, a phase detector **702**, a low pass filter **704**, and a clock signal control unit **706**. The phase detector may refer to one or more logic gates which may be used to measure the difference between the clock signals in some embodiments. The difference may refer to timing difference between the different lower frequency output clock signals in some embodiments. The low pass filter **704** may refer to a filter which passes signals with a frequency lower than a certain threshold in some embodiments. The control unit **706** may refer to a computing circuit which is configured to correct any clock signal spacing error observed within the error correction system in some embodiments. The clock signals “CK_P” and “CK_N” are used as inputs to the quadrature divider **606**. As described above, the quadrature divider **606** produces frequency divided output signals **616**, **618**, **620**, and **622**. As described above, in some embodiments, a clock signal spacing error may be observed in the output clock signals **616**, **618**, **620**, and **622** depending on whether there is skewed timing between the input clock signals “CK_P” and “CK_N”. The clock signal spacing error may refer to the timing between output clock signals **616**, **618**, **620**, and **622** being either above or below a certain time in some embodiments. In other embodiments, a clock signal spacing error may be observed in the output clock signals **616**, **618**, **620**, and **622** if the duty cycle of the input clock signals “CK_P” and “CK_N” is below or above 50%. The clock signal error may be corrected by the error correction

system **700**.

(64) The output clock signals **616**, **618**, **620**, and **622** may be used as an input to the phase detector **702**. The phase detector **702** may include two XOR gates which are configured to receive the output clock signals **616**, **618**, **620**, and **622** and determine the difference between the rising edges of the output clock signals **616**, **618**, **620**, and **622**. An XOR gate may be defined as a digital logic gate that provides a true output (e.g., high or 1) when the number of true inputs is odd. In some embodiments, an XOR gate may function as an inverter which may be activated or deactivated by a switch. In some embodiments, the phase detector **702** may include x-nor gates instead of XOR gates. One of the XOR gates may be configured to measure the difference between output signal **616** and output signal **618** while the second XOR gate measures the difference between output signal **618** and **620**. Further, in some embodiments, the XOR gates of the phase detector **702** may be configured to also measure the difference between output signal **620** and output signal **622** and the difference between output signal **616** and output signal **622**. For example, with reference to FIG. 8, timing diagram **800** showing clock signals associated with a phase detector **702**, is shown according to an exemplary embodiment. Specifically, the timing diagram **800** shows the output of the two XOR gates of the phase detector **702**. The output of the two XOR gates in the phase detector **702** are signals which demonstrate the difference between the rising edge of one clock and the rising edge of the other clock. For example, signal **802** shows the difference between the rising edge the first clock signal **616** and the rising edge of the second clock signal **618**. As another example, the signal **804** shows the difference between the rising edge of the second clock signal **618** and the third clock signal **620**. The signals **802** and **804** demonstrate that no clock signal spacing error was detected by the phase detector **702** because the signals **802** and **804** each have a duty cycle of 50%. In contrast, the output signals of the XOR gates demonstrated in FIG. 9 is a timing diagram which demonstrates that a clock signal spacing error has been detected.

Specifically, the timing diagram **900** shows the output of the two XOR gates of the phase detector **702**. The output of the two XOR gates in the phase detector **702** are signals which demonstrate the difference between the rising edge of one clock and the rising edge of the other clock. For example, signal **902** shows the difference between the rising edge the first clock signal **616** and the rising edge of the second clock signal **618**. As another example, the signal **904** shows the difference between the rising edge of the second clock signal **618** and the third clock signal **620**. The signals **902** and **904** demonstrate that a clock signal spacing error was detected by the phase detector **702** because the signals **902** and **904** each have a duty cycle below or above 50%.

(65) Referring back to FIG. 7, the output signals from the XOR gates are fed into the low pass filter **704** which is configured to measure and amplify the clock signal spacing error as determined by the phase detector **702**. Based on the measured and amplified clock signal spacing error, the low-pass filter may determine control signals “vctrl_p” and “vctrl_n.” A control signal may refer to an electrical or communication signal which may be configured to control the action of an electronic circuit component or device. Measuring the clock signal may refer to determining the amount of clock signal spacing error that is present in the output signals in some embodiments. The amplified clock signal spacing error may refer to a clock signal spacing error measurement that has been proportionally increased in some embodiments. The control signals may then be fed into the clock signal control unit **706**. The clock signal control unit **706** is configured to receive the control signals and generate corrected higher frequency input clock signals “CK_N” and “CK_P” to correct for any clock signal spacing error. The corrected higher frequency input clock signals “CK_N” and “CK_P” may refer to higher frequency input clock signals which have been modified to account for any clock signal spacing error in some embodiments. Specifically, the clock signal control unit **706** may include one or more current-starved inverters **708** and **710** which are used to drive the corrected input clock signals “CK_N” and “CK_P”. The control voltage for the generation of CK_N is the ‘opposite’ (other differential signal) from that that creates CK_P. Thus, the control signals go to separate current-starved inverter paths, and this leads to controllable and

clock signal spacing error free duty cycle. Additional skew correction is possible by including CMOS inverters between CK_P and CK_N (not shown).

(66) Referring now to FIGS. **10A-10B**, control signals for controlling the input clock signals “CK_N” and “CK_P” and the corresponding input clock signals created by the control signals is shown, according to an exemplary embodiment. In FIG. **10A**, the first control signal **1002** and the third control signal **1006** are the control signals generated when no clock signal spacing error is detected by the phase detector **702**. In such a case, the rising edge and/or falling edge of the control signals **1002** and **1006** do not need to be adjusted to account for and correct any clock signal spacing error. When combined, as shown in FIG. **10B**, the voltage control signals **1002** and **1006** create an input clock signal **1104** (e.g., “CK_N” or “CK_P”) which has a standard 50% duty cycle. Referring back to FIG. **10A**, the second clock signal **1004** and the fourth clock signal **1008** are control signals generated when a clock signal spacing error is detected by the phase detector **702**. In such a case, the rising edge and/or the falling edge may be lengthened or shortened to adjust for any clock signal spacing error. When combined, as shown in FIG. **10B**, the voltage control signals **1004** and **1008** create an input clock signal **1108** (e.g., “CK_N” or “CK_P”) which has an adjusted duty cycle (e.g., above or below 50%).

(67) D. Systems for and Methods for Synchronization of Multiple Frequency Dividers

(68) In many analog to digital converters (ADCs), deserializer circuits in link communication (SERDES), or even radio receivers (RF radios), the information is processed synchronously with a clock. As the data rates have increased to well above 20 Gb/s, it is generally preferred to process the data with multiple sub-rate clocks. For example, in one embodiment, a data rate of 20 Gb/s can be captured with two 10 GHz clocks, or four 5 GHz clocks, etc. When using such sub-rate clocks in communication systems with a certain data rate (e.g., 20 Gb/s), it may be desired that the multiple sub-rate clocks are as equally spaced in time as possible so that the timing reference is equal to the full data rate itself. For example, in the case of a 20 Gb/s system, the timing reference would be at 20 GHz, or a period of 50 ps. In such a case, the data rate may be implemented with a sub-rate system with four 5 GHz clocks, where ideally the four clocks' rising edges (timing references) are to be spaced by exactly 50 ps. To ensure this, the sub-rate clocks may be synchronized when dividing the frequency to create the sub-rate clocks. The systems and methods described herein provide a technique for providing such a synchronization. Synchronization may refer to a process for ensuring that the timing of the sub-rate clock signals is matched with the timing of the higher frequency clock signals in some embodiments.

(69) Referring now to FIG. **11**, a schematic diagram showing multiple high-speed clocks **1201**, **1203**, and **1205** which go through frequency division by one or more frequency dividers **1208a-1208c** to generate multiple sub-rate clocks **1202**, **1204**, and **1206**, is shown. For example, high speed clock CK1 may go through a frequency divider (e.g., frequency divider **600**, a fractional divider **212**, and a fractional divider **214**, etc.) to create a sub-rate clock **1202**. “Sub-rate” means that the frequency of the sub-rate clock is less than the data rate itself, and generally is an integer division. The timing diagrams associated with the high-speed clocks **1201**, **1203**, and **1205** and the sub-rate clocks **1202**, **1204**, and **1206** are shown in FIG. **12**. It may be desired that each of the sub-rate clocks **1202**, **1204**, and **1206** are synchronized by ensuring that the frequency dividers are synchronized. The frequency dividers may become unsynchronized because they do not have a known initial state. For example, as shown in FIG. **12**, both of the sub-rate clocks **1302b** may have an initial state of low which matches the initial state of the high frequency clocks **1302a**. Therefore, the sub-rate clocks **1302b** are synchronized. In contrast, one of the sub-rate clocks **1304c** does not have an initial state which matches that of the other sub-rate clock **1304b** or the high frequency clock **1304a**. Therefore, the sub-rate clocks **1304** are not synchronized. The systems and methods described herein provide a way of synchronizing the clocks without the use of an additional high-frequency clock using only the information provided by the sub-rate clocks. Specifically, one sub-rate clock is assigned as the “master” and each sub-rate clock is compared to the master's sub-rate

clock and individually synchronizes itself to the master sub-rate clock which produces a master sub-rate clock signal. A master sub-rate clock signal may refer to a clock signal associated with the master sub-rate clock in some embodiments. A follower sub-rate clock may be a sub-rate clock which is synchronized to the master sub-rate clock. A follower sub-rate clock signal may refer to a clock signal associated with the follower sub-rate clock in some embodiments. Based on the comparison to the master sub-rate clock, a slip divider may be applied to a clock divider of a follower sub-rate clock. The slip divider is explained in more detail below with respect to FIGS. **13A-13C**. For example, in one embodiment, the first sub-rate clock **1202** may be assigned as the “master sub-rate clock.” In such a case, it is the relation between the second sub-rate clock **1204** and the third sub-rate clock **1206** which determines whether the clocks are synchronized or not. In some embodiments, the process for comparing the other sub-rate clocks to the master sub-rate clock may be done independently and simultaneously to save time. Independently and simultaneously synchronizing the sub-rate clocks may refer to multiple sub-rate clocks being synchronized without impacting each other at the same time in some embodiments. The process for comparing the sub-rate clocks to the master sub-rate clocks is explained in more detail with respect to FIGS. **14-17** below.

(70) Referring now to FIG. **13A**, a circuit diagram for a slip divider **1300** is shown, according to an exemplary embodiment. In a typical frequency divider, the rising edge of a high frequency clock (e.g., “CK1,” “CK2,” and/or “CK3”) causes the sub-rate clock to toggle from a high position to a low position or vice versa which causes the position of the sub-rate clock to be directly correlated to the high frequency clocks. The slip divider **1300**, in contrast, includes an additional slip signal and the sub-rate clock toggles based on both the slip signal and the rising edge of the high frequency clocks. A slip signal may refer to a control signal which causes the one or more of the sub-rate clocks to stay low for one or more periods in some embodiments. Specifically, the slip divider **1300** includes an XNOR gate **1310** which includes a slip signal **1306**. The output of the XNOR gate **1310** is used as an input to the flip-flop **1312** which generates the sub-rate clock **1202**. When the slip signal **1306** is low, the XNOR gate **1310** acts like an inverter and the sub-rate clock **1202** toggles as normal and is directly tied to the rising edge of the high frequency clock **1308** (e.g., like in a typical divider). When the slip signal **1306** is high, then the flip-flop **1312**'s input will be the same as the sub-rate clock **1202** and the sub-rate clock **1202** will not toggle on the rising edge of the high frequency clock **1308**. In some embodiments, it may be desired that the slip signal **1306** be synchronized with the high frequency clock **1308** so that the slip signal **1306** masks a desired number of rising edges for the high frequency clock **1308**. If it is determined that a sub-rate clock **1202** is not synchronized based on a comparison with the master clock, then the slip signal **1306** may be activated so that it is high, which synchronizes the sub-rate clock **1202** with the master clock.

(71) Referring now to FIG. **13B-13C**, waveform diagrams **1316** and **1318** including signals associated with the slip divider **1300**, is shown, according to an exemplary embodiment. The waveform diagram **1316** shows the signals associated with the slip divider **1300** when the slip signal **1306** is low. The waveform diagram **1318** shows the signals associated with the slip divider **1300** when the slip signal **1306** is high. As can be seen in the waveform **1316**, the sub-rate clocks **1201** and **1203** are synchronized with the high frequency clocks **1202** and **1204**. Because the clocks are all synchronized, the div_good signal **1320** is set to high in waveform diagram **1316**. The div_good signal **1320** may be described as the control signal which determines when the slip should be activated based on the comparison between different sub-rate clocks and the master sub-rate clock. If the comparison shows that the other sub-rate clocks are synchronized to the master sub-rate clock, then the div_good signal **1320** will be high and the slip signal **1306** will be low. In contrast, if the comparison shows that the other sub-rate clocks are not synchronized to the master sub-rate clock, then the div_good signal **1320** will be low and the slip signal **1306** will be activated to be high. Waveform diagram **1318** shows an exemplary embodiment in which the sub-rate clocks

are not synchronized and the slip signal **1306** is activated to be high. As shown in FIG. **13C**, the div_good signal **1320** is low which indicates that the sub-rate clock **1204** is not synchronized with the sub-rate clock **1202**. Once it has been determined that the sub-rate clocks are not synchronized, a slip signal **1322** may be applied to the sub-rate clock **1204** which causes the sub-rate clock **1204** to not toggle for one rising edge of the master clock.

(72) Referring now to FIG. **14**, a first circuit diagram **1400** of a frequency divider synchronizer is shown, according to an exemplary embodiment. The first circuit diagram **1400** is configured to determine whether one or more follower sub-rate clocks are synchronized to a master sub-rate clock. If the first circuit diagram **1400** determines that the sub-rate clocks are not synchronized, then the first circuit diagram **1400** may be configured to generate a slip signal which is configured to synchronize the follower sub-rate clocks to the master sub-rate clock. The first circuit diagram **1400** determines whether the one or more follower sub-rate clocks are synchronized based on a div_good signal **1405**. Specifically, the first circuit diagram **1400** includes a buffer delay gate **1402** and may refer to a circuit element which is configured to provide a delay, if desired, between the timing of the master sub-rate clock (e.g., sub-rate clock **1202**) and one or more follower sub-rate clocks (e.g., sub-rate clock **1204**) in some embodiments. For example, as shown in FIG. **13B**, the high frequency clock **1203** lags behind the high frequency clock **1201** for a certain time period. Therefore, for the sub-rate clocks to be properly synchronized, the sub-rate clock **1204** should lag behind the sub-rate clock **1202** by the same time period. In such a case, the buffer delay gate **1402** will ensure a proper lag between the sub-rate clocks. In some embodiments, the first circuit diagram **1400** may not include the buffer delay gate **1402** if the high-frequency clocks do not lag behind each other.

(73) The delayed clock signal for the follower sub-rate clock (e.g., clock “ckp2” **1204**) is used as an input to clock a flip flop **1404**. The flip flop **1404** also includes an input of the master sub-rate clock which in this case is ckp1 (e.g., sub-rate clock **1202**). The flip flop **1404** uses the follower sub-rate clock ckp2 (e.g., sub-rate clock **1204**) to sample the master sub-rate clock ckp1. If the second sub-rate clock **1204** lags behind the first sub-rate clock **1202** correctly, the sampling by the flip flop **1404** yields an output of the div_good signal **1405** being high which indicates that the sub-rate clocks are synchronized. If the sub-rate clocks are out of synchronization, then the followers sub-rate clocks' rising edge will align with the master sub-rate clocks being low and the flip flop **1404** will yield the div_good signal **1405** being low which indicates that the clocks are not synchronized. In this way, the flip flop **1404** outputs a logical indication of whether the sub-rate clocks are synchronized.

(74) The div_good signal **1405** is then used as an input to a NOR gate **1406** which is configured to invert the div_good signal **1405**. For example, if the div_good signal **1405** is low, then it is inverted by the NOR gate **1406** to be high. Inversely, if the div_good signal **1405** is high, then it is inverted by the NOR gate to be low. The output of the NOR gate **1406** is fed into flip flop **1408**. In some embodiments, the flip flop **1408** is clocked by the high frequency clock **1203**. This may be done to ensure that the slip signal is only enabled for a desired number of periods of the high frequency clock **1203**. If the div_good signal **1405** is low, which indicates that the sub-rate clocks are not synchronized, then the signal is inverted to produce a high signal. This high signal is fed into flip flop **1408** to synchronize it with the high frequency clock **1203** and produce a to_slip signal **1409** which is high. The to_slip signal **1409** may then be used as an input to AND gate **1414**. Further, the high signal produced by the flip flop **1408** is fed into flip flop **1410** which is also clocked by the high frequency clock to produce a low signal. The low signal is used as an input for the inverter **1412** which produces a high signal which is used as a second input for the AND gate **1414**. Given the high signals which are inputs to the AND gate **1414**, the AND gate **1414** produces pos_det signal **1415** which is fed into the flip-flop **1416**. The flip flop **1416** is configured to clock the pos_det signal **1415** to produce slip signal **1417** which is high. The slip signal **1417** may then be communicated to a control unit associated with the frequency divider which is configured to use

the slip signal **1417** to synchronize the follower sub-rate clocks as described above with respect to FIGS. **13A-13C**. If the div_good signal **1405** is high, then a slip signal **1417** is produced to be low following the logic described above.

(75) Referring now to FIG. **15**, a second circuit diagram **1500** showing an exemplary embodiment of a frequency divider synchronizer is shown, according to an exemplary embodiment. Similarly to the first circuit diagram **1400**, the second circuit diagram **1500** is configured to determine whether one or more follower sub-rate clocks are synchronized to a master sub-rate clock. If the second circuit diagram **1500** determines that the sub-rate clocks are not synchronized, then the second circuit diagram **1500** may be configured to generate a slip signal which is configured to synchronize the follower sub-rate clocks to the master sub-rate clock. The second circuit diagram **1500** determines whether the one or more follower sub-rate clocks are synchronized based on a div_good signal **1505**. The second circuit diagram **1500** includes many of the same elements described above with respect to first circuit diagram **1400**. For example, the second circuit diagram **1500** includes a buffer delay gate **1502** which is similar to buffer delay gate **1402**. The second circuit diagram **1500** also includes a flip flop **1504** which is configured to produce a div_good signal **1505** similarly to flip flop **1404**. Further, the second circuit diagram **1500** includes a NOR gate **1507** with inputs div_good signal **1505** and flush signal **1507** similar to NOR gate **1406** with inputs div_good signal **1405** and flush signal **1507**. The second circuit diagram **1500** also includes flip flops **1508** and **1510** which are similar to flip flops **1408** and **1410** described above. Flip flop **1508** produces to_slip signal **1509** similarly to how flip flop **1408** produces to_slip signal **1409** as described above. The second circuit diagram **1500** includes an inverter **1512** and an AND gate **1514** producing pos_det signal **1515** similarly to the inverter **1412**, the AND gate **1414**, and the pos_det signal **1415** respectively. Similar to pos_det signal **1415**, pos_det signal **1515** is fed into one or more flip flops to produce a slip signal. The second circuit diagram **1500** differs from the first circuit diagram **1400** in that it includes more flip flops than included in the first circuit diagram **1400**. The amount of flip flops used to create the slip signal directly correspond to the amount of periods the slip signal is active for. For example, the first circuit diagram **1400** includes only one flip flop **1416** which creates the slip signal **1417**, which means that the slip signal is only active for one period of the clock **1203**. In contrast, the second circuit diagram **1500** includes two flip flops **1516** and **1518**, which feed into the OR gate **1520** to produce a slip signal **1517**, which is active for two periods of the clock signal **1203**.

(76) In another exemplary embodiment, a circuit diagram may further include a multiplexer which is configured to provide an alternative method of determining the number of periods the slip signal is active. For example, referring now to FIG. **16**, a third circuit diagram **1600** is shown, which is very similar to circuit diagram **1500** but includes a multiplexer **1620**, which is used as an input to the OR gate **1622** to produce the slip signal **1617**. Specifically, if the slip_2 signal **1624** is high, then the circuit diagram **1600** will produce a high slip signal for 2 periods of the clock signal **1203**. However, if the slip_2 signal **1624** is low, then the circuit diagram **1600** will produce a high slip signal for 1 period of the clock signal **1203**.

(77) In another exemplary embodiment, the circuit diagram shown in FIG. **15** can be expanded to include more than two flip flops to create a slip signal which lasts for more than two periods of the clock signal **1203**. For example, referring now to FIG. **17**, a fourth circuit diagram **1700** is shown, according to an exemplary embodiment. The fourth circuit diagram **1700** includes most of the same components as circuit **1500** which are described in more detail above. Further, circuit diagram **1700** includes flip flops **1716**, **1718**, **1720**, and **1722** which are configured to create a slip signal which lasts for more than two periods of the clock signal **1203**. Specifically, the four flip flops signals included in FIG. **17** correspond to keeping a slip signal which is active for four clock periods of the clock signal **1203**. An increase in the amount of flip flops would lead to the same increase in the amount of clock periods the slip signal is active for.

(78) It should be noted that certain passages of this disclosure can reference terms such as “first”

and “second” in connection with device signals, data, inputs, channels, etc., for purposes of identifying or differentiating one from another or from others. These terms are not intended to merely relate entities (e.g., a first input and a second input) temporally or according to a sequence, although in some cases, these entities can include such a relationship. Nor do these terms limit the number of possible entities (e.g., devices) that can operate within a system or environment.

(79) It should be understood that the systems described above can provide multiple ones of any or each of those components. In addition, the systems and methods described above can be provided as one or more computer-readable programs or executable instructions, programmable circuits, or digital logic embodied on or in one or more articles of manufacture. The article of manufacture can be a floppy disk, a hard disk, a CD-ROM, a flash memory card, a PROM, a RAM, a ROM, ASIC or a magnetic tape. In general, the computer-readable programs can be implemented in any programming language, such as LISP, PERL, C, C++, C#, PROLOG, or in any byte code language such as JAVA. The software programs or executable instructions can be stored on or in one or more articles of manufacture as object code.

(80) While the foregoing written description of the methods and systems enables one of ordinary skill to make and use various embodiments of these methods and systems, those of ordinary skill will understand and appreciate the existence of variations, combinations, and equivalents of the specific embodiment, method, and examples herein. The present methods and systems should therefore not be limited by the above described embodiments, methods, and examples, but by all embodiments and methods within the scope and spirit of the disclosure.

Claims

1. A circuit comprising: a frequency divider configured to receive a plurality of first frequency input clock signals and provide a plurality of second frequency output clock signals, wherein the plurality of second frequency output clock signals are lower in frequency than the plurality of first frequency input clock signals; a phase detector configured to determine a difference between the plurality of second frequency output clock signals; a low pass filter configured to measure a clock signal spacing error associated with the plurality of second frequency output clock signals based on the difference between the plurality of second frequency output clock signals and to generate one or more control signals in response to the clock signal spacing error; and a control unit configured to generate one or more corrected first frequency input clock signals based on the one or more control signals, wherein the frequency divider is a quadrature divider.
2. The circuit of claim 1, wherein the quadrature divider comprises a flip flop configured to produce a single second frequency output clock signal of the plurality of second frequency output clock signals having half a frequency of a first frequency input clock signal of the plurality of first frequency input clock signals and one or more latches configured to produce four second frequency output clock signals of the plurality of second frequency output clock signals.
3. The circuit of claim 1, wherein the phase detector includes one or more logic gates configured to receive the plurality of second frequency output clock signals, wherein the plurality of second frequency output clock signals comprise at least a first signal of the plurality of second frequency output clock signals and a second signal of the plurality of second frequency output clock signals and wherein the phase detector compares the first signal to the second signal to detect the clock signal spacing error.
4. The circuit of claim 3, wherein the one or more logic gates are XOR gates.
5. The circuit of claim 1, wherein the low pass filter is configured to amplify the clock signal spacing error and determine the one or more control signals based on the amplified clock signal spacing error.
6. The circuit of claim 1, wherein the clock signal spacing error is a condition in which the timing of the rising edges of the plurality of second frequency output clock signals is not equal between

each of the plurality of second frequency output clock signals.

7. A method comprising: receiving a plurality of first frequency input clock signals; providing a plurality of second frequency output clock signals, wherein the plurality of second frequency output clock signals are lower in frequency than the plurality of first frequency input clock signals, and wherein producing the plurality of second frequency output clock signals is done by a quadrature divider; determining a difference between the plurality of second frequency output clock signals to determine whether a clock signal spacing error is present; measuring the clock signal spacing error; determining one or more control signals based on the clock signal spacing error; and generating a corrected first frequency input clock signal based on the one or more control signals.

8. The method of claim 7, further comprising: receiving the plurality of second frequency output clock signals, wherein the plurality of second frequency output clock signals comprise at least a first signal of the plurality of second frequency output clock signals and a second signal of the plurality of second frequency output clock signals; and comparing the first signal to the second signal to detect the clock signal spacing error.

9. The method of claim 7, further comprising: amplifying the clock signal spacing error; and determining the one or more control signals based on the amplified clock signal spacing error.

10. The method of claim 7, wherein the clock signal spacing error is a condition where a time of a rising edge of a signal of the plurality of second frequency output clock signals is not equal to a time of a rising edge of another signal of the plurality of second frequency output clock signals.

11. The method of claim 7, wherein the quadrature divider comprises a flip flop configured to produce a second frequency output clock signal of the plurality of second frequency output clock signals having half the frequency of the plurality of first frequency input clock signals and one or more latches configured to produce four second frequency output clock signals of the plurality of second frequency output clock signals.

12. The method of claim 7, wherein the clock signal spacing error is a condition in which the timing of the rising edges of the plurality of second frequency output clock signals is not equal between each of the plurality of second frequency output clock signals.

13. A non-transitory computer readable medium storing instructions that when executed by one or more processors cause the one or more processors to: receive a plurality of first frequency input clock signals; provide a plurality of second frequency output clock signals, wherein the plurality of second frequency output clock signals are lower in frequency than the plurality of first frequency input clock signals, wherein producing the plurality of second frequency output clock signals is done by a quadrature divider; determine a difference between the plurality of second frequency output clock signals to determine whether a clock signal spacing error is present; measure the clock signal spacing error; determine one or more control signals based on the clock signal spacing error; and generate one or more corrected first frequency input clock signals based on the one or more control signals.

14. The non-transitory computer readable medium of claim 13, further causing the one or more processors to: receive the plurality of second frequency output clock signals, wherein the plurality of second frequency output clock signals comprise at least a first signal of the plurality of second frequency output clock signals and a second signal of the plurality of second frequency output clock signals; and compare the first signal to the second signal to detect the clock signal spacing error.

15. The non-transitory computer readable medium of claim 13, further causing the one or more processors to: amplify the clock signal spacing error; and determine the one or more control signals based on the amplified clock signal spacing error.

16. The non-transitory computer readable medium of claim 13, wherein the clock signal spacing error is a condition in which a timing of a rising edge of a signal of the plurality of second frequency output clock signals is not equal to a timing of a rising edge of another signal of the

plurality of second frequency output clock signals.

17. The non-transitory computer readable medium of claim 13, wherein the quadrature divider comprises a flip flop configured to produce a second frequency output clock signal of the plurality of second frequency output clock signals having half a frequency of a first frequency input clock signal of the plurality of first frequency input clock signals and one or more latches configured to produce four second frequency output clock signals of the plurality of second frequency output clock signals.
